

Using Display

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Reference: M. Margolis, et al., Arduino Cookbook, 3rd Edition,
Oreilly, 2020

LCD

- Liquid Crystal Display
- LCD displays are further categorized into two types on the basis of the technology used to make them.
 - IPS LCD
 - TFT LCD

TFT LCD

- TFT stands **Thin-film Transistor**
 - de facto, it really **isn't** a type of display.
- TFT is only the technology used to produce LCD display panels.
- TFT LCD displays use an '**Active Matrix Technology**' where the display transistor and capacitor have individual pixels attached to them.

TFT LCD

- Each pixel can have as many as four transistors; for switching them off and on easily.
- TFT displays are widely known for having high contrast ratios, resolution and image quality.
- They are also cheaper to produce but not as cheap as IPS LCD.

IPS LCD

- IPS stands for In-Plane Switching
- It is the most popularly used type of LCD panels for a number of reasons.
 - First, compared to TFT, the crystal/pixel orientation on IPS LCD is different.
 - This modification allows for improved color reproduction, better viewing angles, and reduced energy consumption.
 - This is why IPS LCD is preferred over TFT by most gadgets manufacturers.

LCD

- Generally, LCDs are known as the “backlit displays” because the pixels on the display are powered by a polarized light engineered to the screen.
 - The light passes through the (horizontal and vertical) filters which help determine the pixel’s brightness.
- Backlight
 - makes LCD displays thicker,
 - pixels are generally more closely packed, colours are more natural, and images — sharper.

<https://www.dignited.com/35182/smartphone-display-technologies-explained/>

OLED

- “Organic light-emitting diode”.
- OLED panels produce their own light and do not rely on a backlight.
 - This self-emission is achieved when an electrical current passes through two conductors with an organic carbon-based film between them.

OLED

- For every pixel in an OLED panel, this light emission process takes place, except when black is being displayed, the pixels are turned off instead.
 - This makes OLED the best display that shows true black colour.
- The process of turning off pixels when displaying black is also a battery-wise feature of OLED panels.

OLED

- OLED are generally better at displaying blacks.
- They are also slimmer, dissipate less heat, and possess better contrast ratio when compared to LCDs.
- However, they are more expensive to produce
- Shorter lifespan is also a downside to OLED displays.

AMOLED

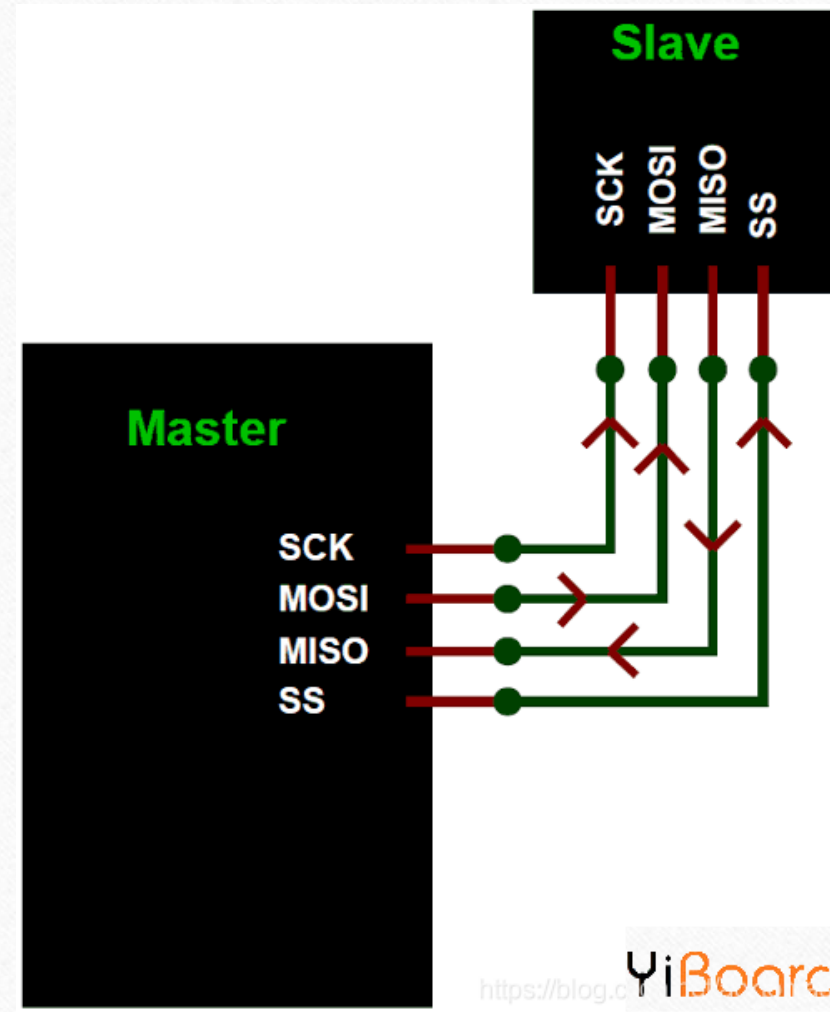
- Active Matrix Organic Light Emitting Diode
- AMOLED is an advanced type of OLED display that uses an “Active Matrix” technology.
- Like OLED, AMOLED pixels also emit their own light and further uses an active matrix system attached to a thin-film transistor (TFT) to exert more control over each pixels.
 - better visual experience; darker blacks, deeper brights, and higher refresh rates.
- AMOLED panels are mostly used in big-sized smartphones as it supports almost any display size.
- One downside to AMOLED panels, though, is poor usability under sunlight.

ILI9341 TFT Module

- The ILI9341 TFT module contains a display controller with the same name: ILI9341.
- It's a color display
- uses **SPI (Serial Peripheral Interface)** interface protocol
- requires 4 or 5 control pins
- The resolution is 240 x 320
 - 76800 pixels.
- works with 3.3V only
 - **doesn't** support 5V (**NOT** 5V tolerant).

<https://simple-circuit.com/interfacing-arduino-ili9341-tft-display/>

- SPI有四條線MISO、MOSI、SS和CLK
 - MISO（主進從出）- 用於向主設備發送數據的從設備線。
 - MOSI（主出從入）- 用於向外設發送數據的主線。
 - SCK（串行時鐘）- 同步主機產生的數據傳輸的時鐘脈衝。
 - SS（從機選擇）- Master可以使用此引腳來啟用和禁用特定設備。



Hardware Required

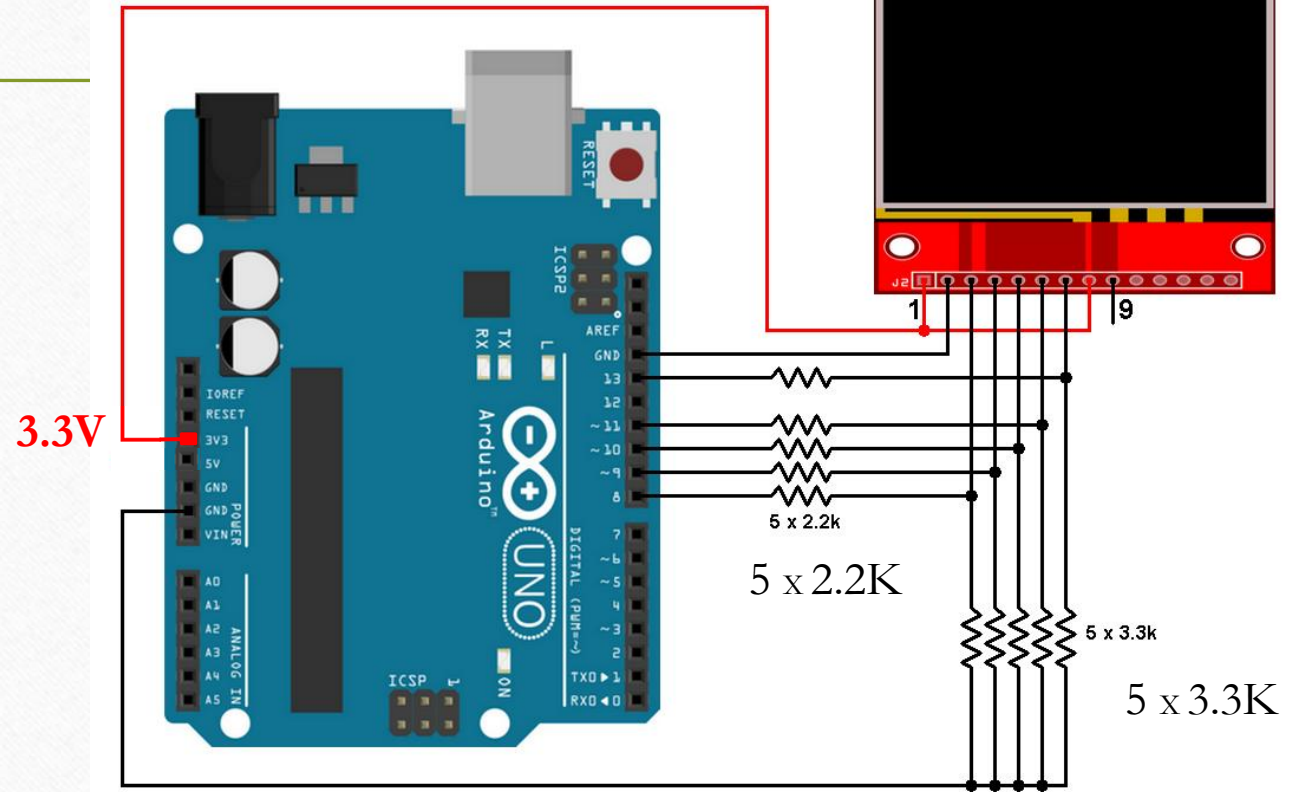
- Arduino UNO board
- ILI9341 TFT display module
- 5 x 3.3k ohm resistor
- 5 x 2.2k ohm resistor
- Breadboard
- Jumper wires

- Interfacing Arduino with ILI9341 TFT display circuit

TFT Pinout:

1	2	3	4	5	6	7	8	9
VCC	GND	CS	RST	D/C	MOSI	SCK	BL	MISO

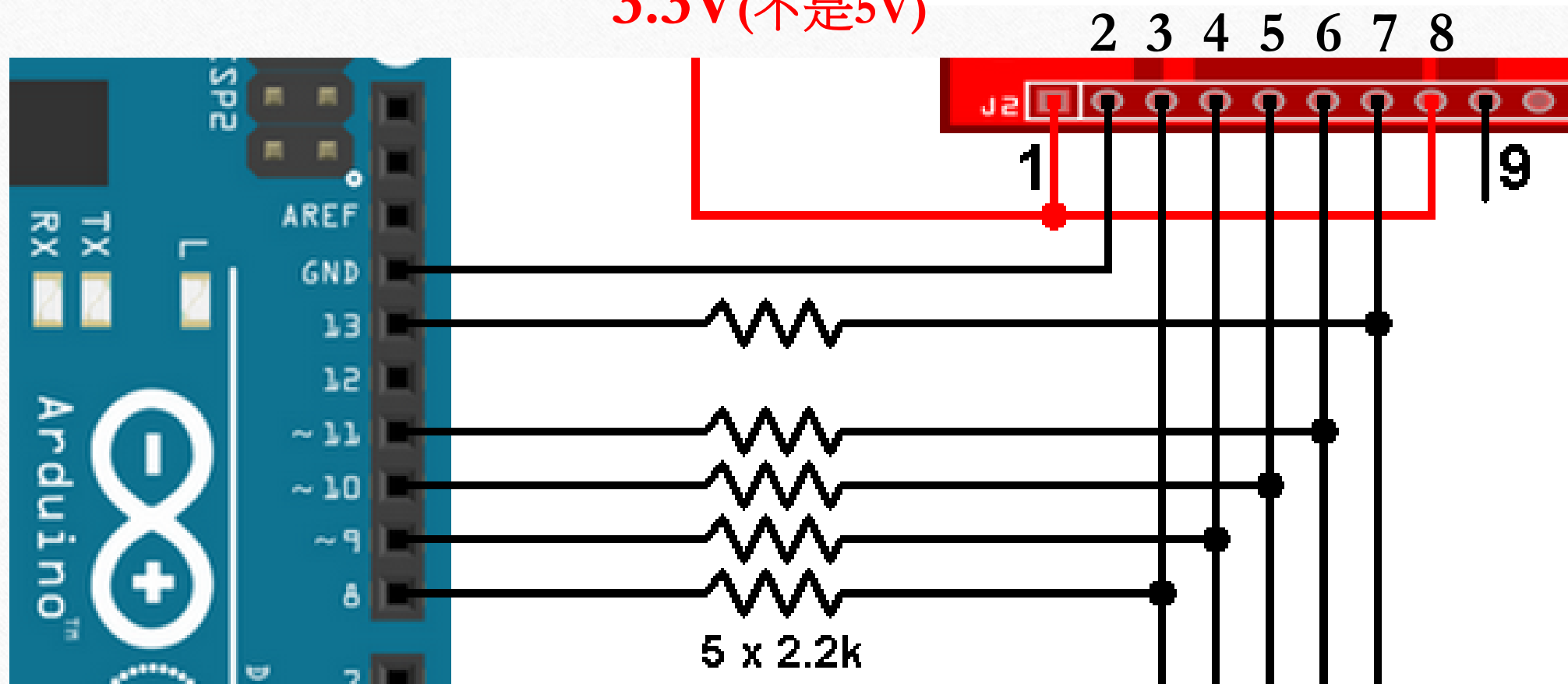
ILI9341 TFT



TFT Pinout:

1	2	3	4	5	6	7	8	9
VCC	GND	CS	RST	D/C	MOSI	SCK	BL	MISO

3.3V(不是5V)



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- 14 pins
 - the first 9 pins are for the display
 - the other 5 pins are for the touch module.
 - Pins numbered from 1 to 9 are (from left to right):
 - VCC (5V), GND (ground),
 - CS (chip select), RST (reset),
 - DC (or D/C: data/command), MOSI (or SDI), SCK (clock), BL (back light LED) and MISO (or SDO).

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- MOSI: master-out slave-in.
SDI: serial data in.
MISO: master-in slave-out.
SDO: serial data out.

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- As mentioned above, the ILI9341 TFT display controller works with **3.3V** only (power supply and control lines).
 - This module has a **built-in 3.3V regulator** which supplies the display controller with 3.3V from the 5V source.
 - The display module is supplied with 5V that comes from the Arduino board.

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- All Arduino UNO board output pins are 5V, connecting a 5V pin to the ILI9341 TFT display may **damage** its controller.
 - To connect the Arduino to the display module, used voltage divider (分壓器) for each line which means there are 5 voltage dividers.
 - Each voltage divider consists of 2.2k and 3.3k resistors, this drops the 5V into 3V which is sufficient.

- CS pin is connected to Arduino digital pin 8,
RST pin is connected to Arduino digital pin 9,
D/C pin is connected to Arduino digital pin 10,

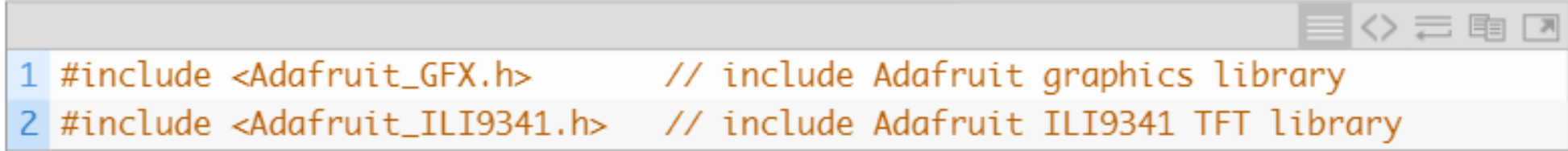
MOSI pin is connected to Arduino digital pin 11,
SCK pin is connected to Arduino digital pin 13.
- VCC pin is connected to Arduino 5V pin,
GND pin is connected to Arduino GND pin,
BL (LED) pin is connected to Arduino 5V pin,
MISO pin is not connected.

Interfacing Arduino with ILI9341 TFT display code

- Requires two libraries from Adafruit Industries:
 - a driver for the ILI9341 TFT display
 - can be installed from Arduino IDE library manager
 - (Sketch —> Include Library —> Manage Libraries ..., in the search box write “ili9341” and choose the one from Adafruit).
 - Adafruit graphics library
 - can be installed from Arduino IDE library manager.

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- The previous two libraries can also be installed manually:
Download both libraries from the following two links:
[Adafruit ILI9341 TFT library](#) —> [direct link](#)
[Adafruit graphics library](#) —> [direct link](#)
 - Go to Arduino IDE —> Sketch —> Include Library —> Add .ZIP Library ... and browse for the .zip file (previously downloaded).
The same thing for the second file.

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- The previous 2 libraries are included in the main code as shown below:

A screenshot of a code editor window with a light gray background and a dark gray title bar. The title bar contains several icons: a hamburger menu, a code symbol, a refresh symbol, a document symbol, and a window symbol. The code is displayed in a monospaced font with orange text. Line numbers 1 and 2 are shown in a light blue box on the left. The code consists of two lines of preprocessor directives.

```
1 #include <Adafruit_GFX.h>      // include Adafruit graphics library
2 #include <Adafruit_ILI9341.h>  // include Adafruit ILI9341 TFT library
```


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- The ILI9341 TFT display is connected to Arduino hardware SPI module pins (clock and data)
 - the other pins are defined as shown below
 - CS (chip select), RST (reset) and DC (data/command) :

```
1 #define TFT_CS    8      // TFT CS  pin is connected to arduino pin 8
2 #define TFT_RST   9      // TFT RST pin is connected to arduino pin 9
3 #define TFT_DC    10     // TFT DC  pin is connected to arduino pin 10
```

Full Arduino code

```
#include <Adafruit_GFX.h>    // include Adafruit graphics library
#include <Adafruit_ILI9341.h> // include Adafruit ILI9341 TFT library

#define TFT_CS  8    // TFT CS pin is connected to arduino pin 8
#define TFT_RST 9    // TFT RST pin is connected to arduino pin 9
#define TFT_DC  10   // TFT DC pin is connected to arduino pin 10
// initialize ILI9341 TFT library
Adafruit_ILI9341 tft = Adafruit_ILI9341(TFT_CS, TFT_DC, TFT_RST);
```

```
void setup() {  
  Serial.begin(9600);  
  Serial.println("ILI9341 Test!");
```

```
  tft.begin();
```

```
  // read diagnostics (optional but can help debug problems)  
  uint8_t x = tft.readcommand8(ILI9341_RDMODE);  
  Serial.print("Display Power Mode: 0x"); Serial.println(x, HEX);  
}
```

```
void loop(void) {
```

```
}
```